AquaSentinel: value-engineered production cost ≈ ₹48,000 (7 Nov 2026)

≈ ₹47,900
per sentinel at 100 units (ex-GST)

≈ ₹64,600
first prototype (ex-GST)

−47 %
from ₹90,800 before value engineering

73–84
sentinels for the cost of one ₹35–40 lakh station

≈ 12 months
payback at ₹8,000 a month

Target set by the team: a sentinel at about ₹50,000. Starting point: the Raspberry Pi version of R3 at ₹1,18,580 (prototype) / ₹90,800 (100 units), from `02_Pi_Based_Product_Cost_Estimate_2026-11-06.md` and the researched costed BOM (23 Oct).

How the cost falls: the parts bolted to the bridge move from 316L stainless to galvanised steel; welded and anodised aluminium parts become bolted or HDPE parts; the wiper and the cool-box sampler are dropped (the Pi sensor set does not use them); fasteners, assembly and freight are scaled for 100 units. **Float mass is unchanged** (steel replaces stainless of the same density; HDPE parts are lighter), so the buoyancy and stability checks still hold. The guide pole keeps its 2-inch section; galvanised pipe steel yields at ≥ 210 MPa against 170 MPa for 316L, so the quoted pole safety factor (2.66) stays conservative.

Group	Item	At 100 before (₹)	Prototype (₹)	At 100 units (₹)	Change
Bridge kit	Guide pole, 2-inch	7,253	1,529	1,479	316L stainless → heavy-class GI pipe (same section; yield ≥ 210 MPa vs 170 MPa)
Bridge kit	Pier brackets (2)	9,708	3,251	2,601	316L TIG weldment → hot-dip galvanised MS, MIG welded
Bridge kit	Pole clamp caps (2) + stage target	2,830	1,171	871	316L → galvanised MS
Bridge kit	Stop collars, top cap, buffers	1,350	1,350	1,250	unchanged (bought)
Bridge kit	Anchors + injection mortar	2,460	1,500	1,380	A4 stainless studs → HDG studs
Float	Twin hulls, foam-filled	2,988	9,868	2,988	unchanged: rotomoulded PE at volume (prototype: welded PE sheet)
Float	Crossbeams (4)	2,970	1,796	1,652	aluminium SHS kept; anodising dropped
Float	Equipment deck	3,305	1,000	900	laser-cut aluminium tread plate → 10 mm HDPE marine board
Float	Guide carriage (fixed half + gate)	8,750	3,200	2,400	TIG-welded laser-cut weldments → bolted aluminium angle
Float	Guide rollers UHMW-PE (8)	1,492	2,233	1,492	unchanged
Float	Hinge + latch pins, struts	1,440	1,300	1,130	struts 316L → GI tube; pins stay stainless
Float	Sensor well: sleeve, hatch ring, guard cage, deflector	4,129	1,940	1,567	hatch ring 316L → HDPE; cage 316L → galvanised MS
Float	Sensor cartridge (cap, spine, spiders, handle)	2,955	1,300	1,030	machined acetal + wiper drive → turned HDPE cap; wiper removed (Pi set has no wiper)

Group	Item	At 100 before (₹)	Prototype (₹)	At 100 units (₹)	Change
Float	Wiper arm + brushes	447	0	0	removed (not used by the Raspberry Pi sensor set)
Float	Pod rails/plate, canopy, mast, handles	4,925	1,900	1,520	aluminium/316L → HDPE board, aluminium angle, GI conduit; 4 handles → 2
Float	Sampler strainer + bottle rack	1,131	900	734	carousel cassette → simple 4-bottle HDPE rack
Electronics	Pi 3B+, pH, TDS, turbidity, DS18B20, DHT11, MQ-3, rain, LDR, SIM800L, pump, driver, LED	–	7,189	6,111	team hardware at retail; −15 % volume
Electronics	2 × ADS1115, RTC, bucks, passives, SD card, buffers	–	2,542	2,161	bench additions
Electronics	Wake-up timer	–	555	472	₹3,000 timer board → TPL5110 timer module
Electronics	Enclosure, glands, harness, antenna, beacon LEDs	–	2,610	2,480	IP67 PC box + strobes → IP65 ABS box + 12 V LED modules
Power	20 W solar, 12.8 V 6 Ah LiFePO4, MPPT, fuse	2,758	3,322	2,758	unchanged
Sampler	Bottles, caps, tubing	650	760	650	cool-box cassette dropped
Fasteners	Bolts, roller axles, lifting eyes, bought mech.	4,680	4,400	3,460	A4 stainless → A2 + HDG; 4 lifting eyes → 2
Assembly	Assembly, leak test + calibration, freight, mould share	10,326	9,000	6,800	jigged bolted assembly; rotomould ₹2.5 lakh over 100 units
Total		**90,800**	**64,616**	**47,886**	

- **Per sentinel in production (100 units): ≈ ₹47,886 ex-GST ≈ ₹56,505 with GST.** Bridge kit ₹7,581, float ₹40,305.
- First prototype (1 unit): ≈ ₹64,616 ex-GST.
- One real-time station (₹35–40 lakh, CSE) = about **73–84 sentinels**.

Service model at ₹8,000 a month

Cost to serve one float a year: ₹57,177 (hardware over 5 years ₹9,577; installation ₹3,000; monthly visit ₹18,000; probes and buffers ₹9,000; SIM ₹2,400; alerts ₹1,200; 4 lab samples ₹14,000). Fee ₹96,000 a year → **margin ≈ 40 %**, hardware paid back in **≈ 11.9 months**.

Rates used and their basis

- Galvanised steel fabrication ₹100/kg: GI/MS stock ₹70–85/kg plus hot-dip galvanising ₹16–35/kg (Indian supplier listings, Nov 2026). 2-inch heavy GI pipe ₹80/kg.
- TPL5110 timer module ≈ ₹555 ex-GST (Evelta); team hardware at retail prices from the 6 Nov estimate.
- Everything else from the researched costed BOM (23 Oct). **These are estimates: get 3 supplier quotes before quoting a customer.**

Trade-offs to state honestly

- Galvanised steel needs an annual coating check (stainless did not). Fasteners in water stay stainless.
- The CAD and drawings still show 316L; this is the production cost-down plan, to be modelled in SOLIDWORKS before the pilot.
- Hobby-grade sensors, no dissolved oxygen, 2G SMS: alerts rely on sudden change and monthly calibration.
